

Jensen's Formula and the Mahler Measure as Pandrosion Quotient Integrals

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Abstract

We show that Jensen's formula, the argument principle, and the Mahler measure admit natural Pandrosion decompositions. The argument principle becomes a *partition* of the winding number: each root α_k contributes 0 or 1 via the contour integral of its Pandrosion force $F_k = Q(\alpha_k, z)/P(z)$. Jensen's formula decomposes per-quotient: $\int \log |Q(\alpha_k, \cdot)| d\theta/(2\pi) = \sum_{j \neq k} \log^+ |\alpha_j|$. The Mahler measure satisfies the *Pandrosion Mahler identity* $m(P) = \log^+ |\alpha_k| + m(Q_k)$, expressing $m(P)$ as the contribution of any single root plus the Mahler measure of its Pandrosion quotient.

1 Root counting via Pandrosion forces

In the electrostatic picture (Paper 26), the force from root α_k at the point z is $F_k(z) = Q(\alpha_k, z)/P(z) = 1/(z - \alpha_k)$. The argument principle integrates the total force around a circle:

Theorem 1.1 (Pandrosion root-counting formula). *Let Γ_R be the circle $|z| = R$ traversed counterclockwise. Then:*

$$n(R) = \frac{1}{2\pi i} \oint_{\Gamma_R} \frac{P'(z)}{P(z)} dz = \sum_{k=1}^d \frac{1}{2\pi i} \oint_{\Gamma_R} \frac{Q(\alpha_k, z)}{P(z)} dz = \sum_{k=1}^d \mathbf{1}_{|\alpha_k| < R}, \quad (1)$$

where $n(R)$ is the number of roots of P inside $|z| < R$.

Proof. The total field decomposes: $P'/P = \sum_k Q(\alpha_k, z)/P(z) = \sum_k 1/(z - \alpha_k)$. Each integral $(2\pi i)^{-1} \oint_{\Gamma_R} (z - \alpha_k)^{-1} dz$ equals 1 if $|\alpha_k| < R$ and 0 if $|\alpha_k| > R$, by Cauchy's integral formula. $\square$

Remark 1.2. This is a *partition of the winding number*: the integer $n(R)$ is decomposed into binary contributions, one per root, each determined by the contour integral of the corresponding Pandrosion force.

2 Jensen's formula per-quotient

Theorem 2.1 (Jensen's formula). *For monic $P(z) = \prod_{k=1}^d (z - \alpha_k)$:*

$$\frac{1}{2\pi} \int_0^{2\pi} \log |P(Re^{i\theta})| d\theta = \sum_{k=1}^d \log \max(R, |\alpha_k|). \quad (2)$$

Since $\log |P| = \sum_k \log |z - \alpha_k| = \log |z - \alpha_k| + \log |Q(\alpha_k, z)|$, we obtain:

Corollary 2.2 (Jensen per-quotient).

$$\frac{1}{2\pi} \int_0^{2\pi} \log |Q(\alpha_k, Re^{i\theta})| d\theta = \sum_{\substack{j=1 \\ j \neq k}}^d \log \max(R, |\alpha_j|). \quad (3)$$

The circle integral of $\log |Q(\alpha_k, \cdot)|$ measures the “external complexity” of P at radius R , excluding root α_k .

Proof. $\log |P| = \log |z - \alpha_k| + \log |Q(\alpha_k, z)|$. Integrate:

$$\sum_j \log \max(R, |\alpha_j|) = \log \max(R, |\alpha_k|) + \int \log |Q| \frac{d\theta}{2\pi}.$$

Subtract $\log \max(R, |\alpha_k|)$ from both sides. □

3 The Pandrosion Mahler identity

Definition 3.1 (Mahler measure). The **logarithmic Mahler measure** of a monic polynomial P is:

$$m(P) := \frac{1}{2\pi} \int_0^{2\pi} \log |P(e^{i\theta})| d\theta = \sum_{k=1}^d \log^+ |\alpha_k|,$$

where $\log^+(x) = \max(0, \log x)$.

Theorem 3.2 (Pandrosion Mahler identity). *For every root α_k :*

$$\boxed{m(P) = \log^+ |\alpha_k| + m(Q_k)}, \quad (4)$$

where $Q_k(z) := Q(\alpha_k, z) = \prod_{j \neq k} (z - \alpha_j)$.

The Mahler measure of P decomposes into:

- the contribution of root α_k (zero if inside the unit disk), plus
- the Mahler measure of the Pandrosion quotient Q_k (contribution of all other roots).

Proof. Set $R = 1$ in Corollary 2.2: $m(Q_k) = \sum_{j \neq k} \log^+ |\alpha_j| = m(P) - \log^+ |\alpha_k|$. □

Remark 3.3 (Connection to Paper 10). Paper 10 used the Mahler measure in the context of Lehmer’s conjecture. The identity $m(P) = \log^+ |\alpha_k| + m(Q_k)$ gives a *recursive* formula: strip one root at a time via the Pandrosion quotient, each step reducing the Mahler measure by $\log^+ |\alpha_k|$. After d steps, $m = 0$ (the empty product).

4 L^2 norms of Pandrosion quotients

Proposition 4.1 (Parseval identity for quotients). *On the unit circle:*

$$\frac{1}{2\pi} \int_0^{2\pi} |Q(\alpha_k, e^{i\theta})|^2 d\theta = \sum_{n=0}^{d-1} |c_n^{(k)}|^2, \quad (5)$$

where $Q_k(z) = \sum_{n=0}^{d-1} c_n^{(k)} z^n$.

The L^2 norm of Q_k on the unit circle is the **Bombieri norm** of the $(d - 1)$ -st degree polynomial obtained by removing root α_k .

5 The Pandrosion integral dictionary

Classical formula	Pandrosion form	Result
Argument principle	$n(R) = \sum_k \oint F_k dz / (2\pi i)$	Root count
Jensen's formula	$\int \log Q_k d\theta / (2\pi) = \sum_{j \neq k} \log^+ \alpha_j $	Per-quotient
Mahler measure	$m(P) = \log^+ \alpha_k + m(Q_k)$	Recursive
L^2 norm	$\ Q_k\ _{L^2}^2 = \sum c_n^{(k)} ^2$	Parseval

Jensen's formula, the argument principle, and the Mahler measure all decompose naturally along the Pandrosion quotients. Each root α_k contributes individually through $Q_k = P/(z - \alpha_k)$, and the global invariants of P are sums (or products) of quotient-level invariants. This is the *integral* counterpart of the algebraic dictionary established in Papers 14–22.

References

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